

PSEUDOCODE FOR:

Surface Blasting Design Calculator

BEGIN

INPUT Rock_Type

INPUT Blasting_Diameter (BD)

INPUT Average_In_Hole_Density (ρ)

INPUT Spacing_to_Burden_Ratio (SB)

INPUT Bench_Height (BH)

IF Rock_Type = "hard" THEN

$K \leftarrow 0.70$

ELSE IF Rock_Type = "medium" THEN

$K \leftarrow 0.45$

ELSE IF Rock_Type = "soft" THEN

$K \leftarrow 0.30$

ELSE IF Rock_Type = "very soft" THEN

$K \leftarrow 0.25$

ELSE

 DISPLAY "Invalid Rock Type"

 STOP

END IF

$Mc \leftarrow (\rho \times BD^2) \div 1273$

$SL \leftarrow (30 \times BD) \div 1000$

$Burden \leftarrow \sqrt{((Mc \times (BH - SL)) \div (SB \times K \times BH))}$

Spacing $\leftarrow$ Burden // Since SB = 1

$SD \leftarrow Burden \div 3$

$K_{act} \leftarrow ((BH - SL + SD) \times Mc) \div (Burden \times Spacing \times BH)$

DISPLAY "Stemming Length =", SL

DISPLAY "Burden =", Burden

DISPLAY "Actual Powder Factor =", K_{act}

END

Fragmentation Calculator

BEGIN

DISPLAY "Enter rock type (hard, medium, soft, very soft)"

INPUT rock

CONVERT rock TO lowercase

IF rock = "hard" THEN

$A \leftarrow 10$

$K \leftarrow 0.70$

ELSE IF rock = "medium" THEN

$A \leftarrow 8$

$K \leftarrow 0.45$

ELSE IF rock = "soft" THEN

$A \leftarrow 6$

$K \leftarrow 0.30$

ELSE IF rock = "very soft" THEN

$A \leftarrow 4$

$K \leftarrow 0.25$

ELSE

DISPLAY "Invalid rock type."

STOP

END IF

DISPLAY "Rock Factor (A) = ", A

DISPLAY "Powder Factor (K) = ", K

DISPLAY "Enter Relative Effective Energy (E):"

INPUT E

DISPLAY "Enter Average In-Hole Density (ρ):"

INPUT ρ

DISPLAY "Enter Bench Height (BH):"

INPUT BH

DISPLAY "Enter Stemming Length (SL):"

INPUT SL

$M_c \leftarrow (\rho \times 200^2) \div 1273$

$CL \leftarrow BH - SL$

$Q \leftarrow M_c \times CL$

$X_{50} \leftarrow A \times K^{(-0.8)} \times Q^{(1/6)} \times (115 \div E)^{(19/20)}$

DISPLAY "Mean Fragmentation = ", X_{50}

END